Администрирование локальных сетей

Настройка VPN

Пакавира Арсениу Висенте Луиш

1 июля 2026

Российский университет дружбы народов, Москва, Россия

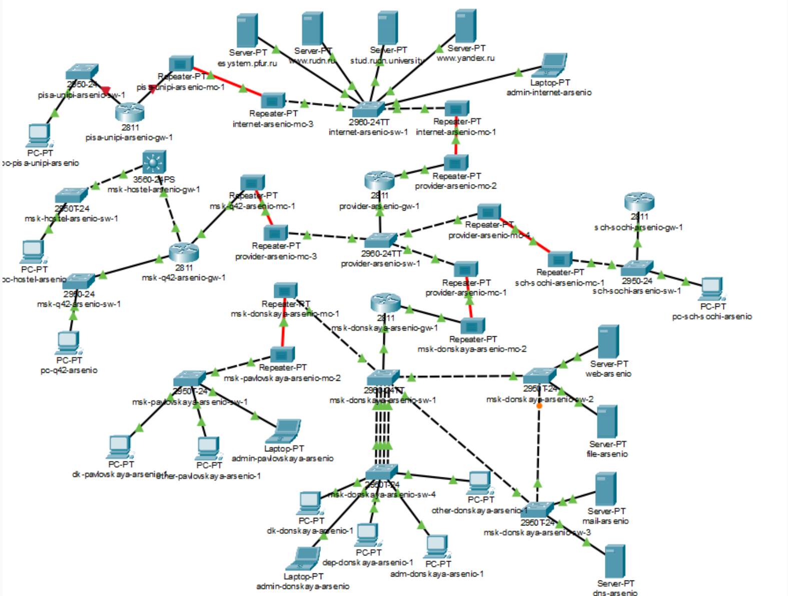
Цели и задачи работы

Цель лабораторной работы

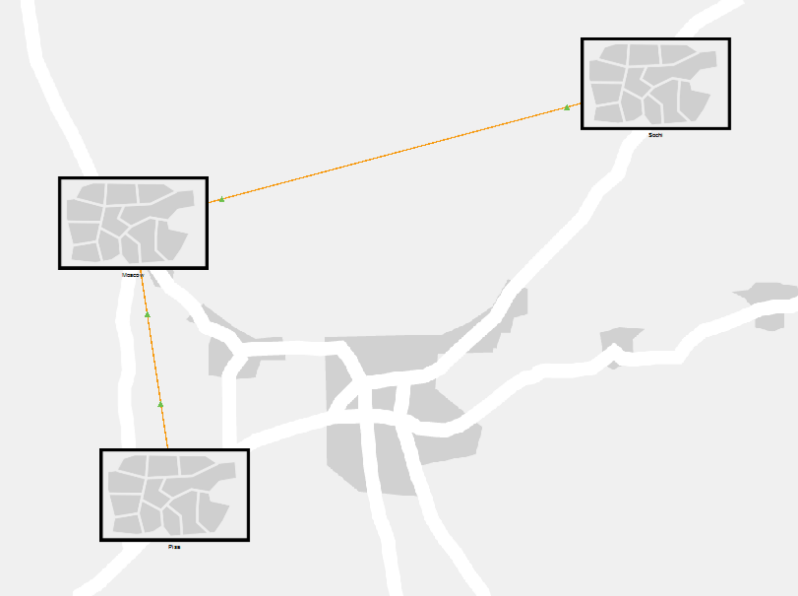
Получить навыки настройки VPN-туннеля через незащищённое Интернет-соединение.

Размещение сети Университета г. Пиза

Общая логическая схема сети



Физическая рабочая область



Здание Университета г. Пиза

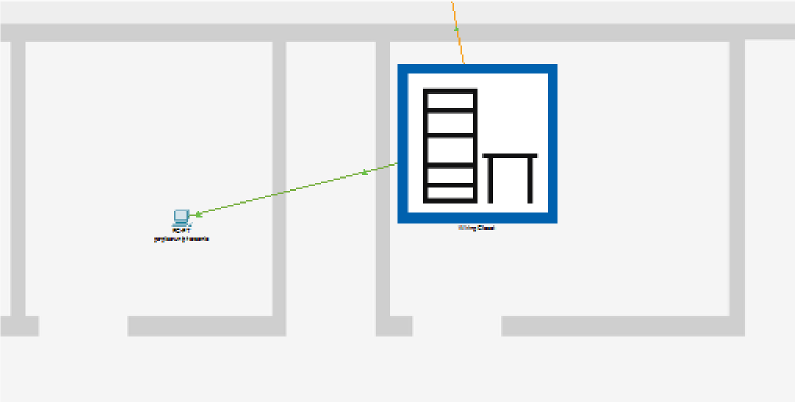
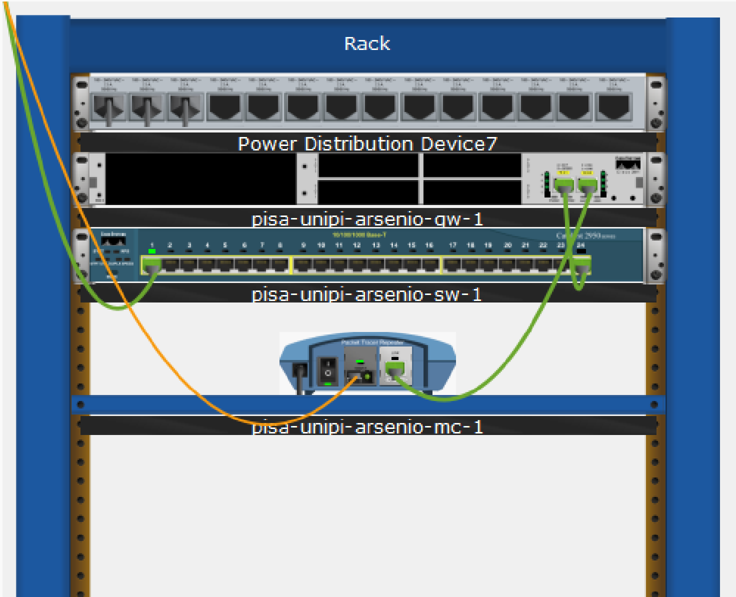


Рис. 3: Размещение здания Университета г. Пиза в физической области

Оборудование в стойке



Первоначальная настройка оборудования

Настройка маршрутизатора Пизы

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#host pisa-unipi-arsenio-gw-1
pisa-unipi-arsenio-gw-1(config)#line vty 0 4
pisa-unipi-arsenio-gw-1(config-line)#pass cisco
pisa-unipi-arsenio-gw-1(config-line)#login
pisa-unipi-arsenio-gw-1(config-line)#line cons 0
pisa-unipi-arsenio-gw-1(config-line)#pass cisco
pisa-unipi-arsenio-gw-1(config-line)#login
pisa-unipi-arsenio-gw-1(config-line)#enab sec cisco
pisa-unipi-arsenio-gw-1(config)#serv pass
pisa-unipi-arsenio-gw-1(config)#user admin priv 1 sec cisco
pisa-unipi-arsenio-gw-1(config)#ip domain-name unipi.edu
pisa-unipi-arsenio-gw-1(config)#crypto key gen rsa
The name for the keys will be: pisa-unipi-arsenio-gw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-arsenio-gw-1(config)#line vty 0 4
*Mar 1 0:6:28.262: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:6:28.262: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-unipi-arsenio-gw-1(config-line)#trans in ssh
pisa-unipi-arsenio-gw-1(config-line)#
pisa-unipi-arsenio-gw-1(config-line)#int f0/0
pisa-unipi-arsenio-gw-1(config-if)#no sh

pisa-unipi-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

Настройка интерфейсов и GRE-туннеля

```
pisa-unipi-arsenio-gw-1(config-if)#int f0/0.401
pisa-unipi-arsenio-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

pisa-unipi-arsenio-gw-1(config-subif)#enc dot1q 401
pisa-unipi-arsenio-gw-1(config-subif)#ip addr 10.131.0.1 255.255.255.0
pisa-unipi-arsenio-gw-1(config-subif)#descr unipi-main
pisa-unipi-arsenio-gw-1(config-subif)#int f0/1
pisa-unipi-arsenio-gw-1(config-if)#no sh

pisa-unipi-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

pisa-unipi-arsenio-gw-1(config-if)#ip addr 192.0.2.20 255.255.255.0
pisa-unipi-arsenio-gw-1(config-if)#descr internet
pisa-unipi-arsenio-gw-1(config-if)#exit
pisa-unipi-arsenio-gw-1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
pisa-unipi-arsenio-gw-1(config)#
pisa-unipi-arsenio-gw-1(config)#int tun 0

pisa-unipi-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

pisa-unipi-arsenio-gw-1(config-if)#ip addr 10.128.255.254 255.255.255.252
pisa-unipi-arsenio-gw-1(config-if)#tun sou f0/1
pisa-unipi-arsenio-gw-1(config-if)#tun dest 198.51.100.2
pisa-unipi-arsenio-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

pisa-unipi-arsenio-gw-1(config-if)#int loop 0

pisa-unipi-arsenio-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

pisa-unipi-arsenio-gw-1(config-if)#op addr 10.128.254.5 255.255.255.255
                                     ^
% Invalid input detected at '^' marker.

pisa-unipi-arsenio-gw-1(config-if)#ip addr 10.128.254.5 255.255.255.255
pisa-unipi-arsenio-gw-1(config-if)#exit
pisa-unipi-arsenio-gw-1(config)#ip route 10.128.254.1 255.255.255.255 10.128.255.253
pisa-unipi-arsenio-gw-1(config)#router ospf 1
pisa-unipi-arsenio-gw-1(config-router)#router-id 10.128.254.5
pisa-unipi-arsenio-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
```

Настройка коммутатора Пизы

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#host pisa-unipi-arsenio-sw-1
pisa-unipi-arsenio-sw-1(config)#line vty 0 4
pisa-unipi-arsenio-sw-1(config-line)#pass cisco
pisa-unipi-arsenio-sw-1(config-line)#login
pisa-unipi-arsenio-sw-1(config-line)#line cons 0
pisa-unipi-arsenio-sw-1(config-line)#pass cisco
pisa-unipi-arsenio-sw-1(config-line)#login
pisa-unipi-arsenio-sw-1(config-line)#enab sec cisco
pisa-unipi-arsenio-sw-1(config)#serv pass
pisa-unipi-arsenio-sw-1(config)#user admin priv 1 sec cisco
pisa-unipi-arsenio-sw-1(config)#ip domain-name unipi.edu
pisa-unipi-arsenio-sw-1(config)#crypto key gen rsa
The name for the keys will be: pisa-unipi-arsenio-sw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-arsenio-sw-1(config)#line vty 0 4
*Mar 1 0:12:42.387: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:12:42.387: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-unipi-arsenio-sw-1(config-line)#transport in ssh
pisa-unipi-arsenio-sw-1(config-line)#
pisa-unipi-arsenio-sw-1(config-line)#int f0/24
pisa-unipi-arsenio-sw-1(config-if)#sw mode trunk

pisa-unipi-arsenio-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

pisa-unipi-arsenio-sw-1(config-if)#int f0/1
pisa-unipi-arsenio-sw-1(config-if)#sw mode acc
pisa-unipi-arsenio-sw-1(config-if)#sw acc vlan 401
% Access VLAN does not exist. Creating vlan 401
pisa-unipi-arsenio-sw-1(config-if)#vlan 401
pisa-unipi-arsenio-sw-1(config-vlan)#name unipi-main
pisa-unipi-arsenio-sw-1(config-vlan)#int vlan 401
pisa-unipi-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

pisa-unipi-arsenio-sw-1(config-if)#no sh
```

Настройка оконечного устройства

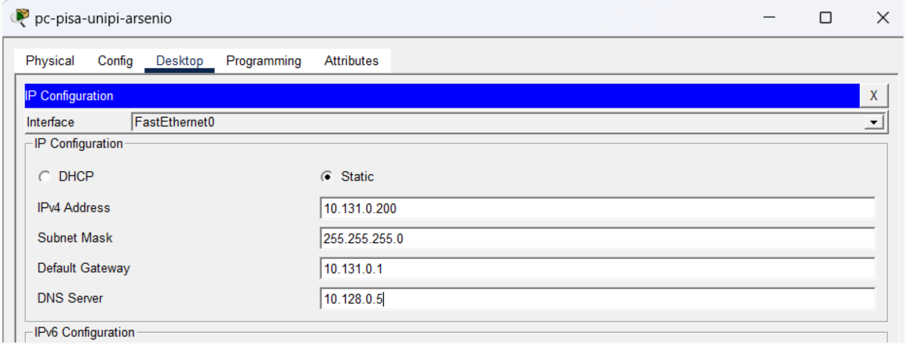


Рис. 8: Настройка статической IPv4-адресации на pc-pisa-unipi-arsenio

Настройка GRE VPN

GRE-туннель на маршрутизаторе «Донская»

```
msk-donskaya-arsenio-gw-1>en
Password:
msk-donskaya-arsenio-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-arsenio-gw-1(config)#int tun 0

msk-donskaya-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

msk-donskaya-arsenio-gw-1(config-if)#ip addr 10.128.255.253 255.255.255.252
msk-donskaya-arsenio-gw-1(config-if)#tun sou f0/1.4
msk-donskaya-arsenio-gw-1(config-if)#tun dest 192.0.2.20
msk-donskaya-arsenio-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

msk-donskaya-arsenio-gw-1(config-if)#int loop 0

msk-donskaya-arsenio-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

msk-donskaya-arsenio-gw-1(config-if)#ip addr 10.128.254.
00:38:49: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.5 on Tunnel0 from LOADING to FULL, Loading Done
1 255.255.255.255
msk-donskaya-arsenio-gw-1(config-if)#
msk-donskaya-arsenio-gw-1(config-if)#exit
msk-donskaya-arsenio-gw-1(config)#ip route 10.128.254.5 255.255.255.255 10.128.255.254
msk-donskaya-arsenio-gw-1(config)#^Z
msk-donskaya-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-gw-1#
```

Проверка доступности узла Пизы

```
C:\>
C:\>ping 10.131.0.200

Pinging 10.131.0.200 with 32 bytes of data:

Request timed out.
Reply from 10.131.0.200: bytes=32 time<1ms TTL=126
Reply from 10.131.0.200: bytes=32 time=1ms TTL=126
Reply from 10.131.0.200: bytes=32 time=10ms TTL=126

Ping statistics for 10.131.0.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>tracert 10.131.0.200

Tracing route to 10.131.0.200 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.128.6.1
  2   1 ms      10 ms     10 ms     10.128.255.254
  3   2 ms      10 ms     10 ms     10.131.0.200

Trace complete.

C:\>
```

Состояние Tunnel0 на маршрутизаторе «Донская»

```
msk-donskaya-arsenio-gw-1#
msk-donskaya-arsenio-gw-1#sh int tun 0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 10.128.255.253/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 198.51.100.2 (FastEthernet0/1.4), destination 192.0.2.20
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 48 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     32 packets input, 2436 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
      0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out

msk-donskaya-arsenio-gw-1#
```

Таблица маршрутизации сети «Донская»

```
msk-donskaya-arsenio-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 29 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
L       10.128.6.1/32 is directly connected, FastEthernet0/0.104
C       10.128.254.1/32 is directly connected, Loopback0
S       10.128.254.5/32 [1/0] via 10.128.255.254
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.1/32 is directly connected, FastEthernet0/1.5
C       10.128.255.4/30 is directly connected, FastEthernet0/1.6
L       10.128.255.5/32 is directly connected, FastEthernet0/1.6
O       10.128.255.8/30 [110/2] via 10.128.255.2, 00:28:17, FastEthernet0/1.5
                        [110/2] via 10.128.255.6, 00:28:17, FastEthernet0/1.6
C       10.128.255.252/30 is directly connected, Tunnel0
L       10.128.255.253/32 is directly connected, Tunnel0
S       10.129.0.0/16 [1/0] via 10.128.255.2
O       10.129.0.0/24 [110/2] via 10.128.255.2, 00:32:47, FastEthernet0/1.5
O       10.129.1.0/24 [110/2] via 10.128.255.2, 00:32:47, FastEthernet0/1.5
O       10.129.128.0/24 [110/3] via 10.128.255.2, 00:32:47, FastEthernet0/1.5
S       10.130.0.0/16 [1/0] via 10.128.255.6
O       10.130.0.0/24 [110/2] via 10.128.255.6, 00:28:17, FastEthernet0/1.6
O       10.130.1.0/24 [110/2] via 10.128.255.6, 00:28:17, FastEthernet0/1.6
O       10.131.0.0/24 [110/1001] via 10.128.255.254, 00:02:05, Tunnel0
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/28 is directly connected, FastEthernet0/1.4
L       198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*   0.0.0.0/0 [1/0] via 198.51.100.1

msk-donskaya-arsenio-gw-1#
```

Состояние Tunnel0 на маршрутизаторе Пизы

```
pisa-unipi-arsenio-gw-1#sh int tun 0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 10.128.255.254/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 192.0.2.20 (FastEthernet0/1), destination 198.51.100.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 60 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     37 packets input, 3412 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
      0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out

pisa-unipi-arsenio-gw-1#
```

Таблица маршрутизации маршрутизатора Пизы

```
pisa-unipi-arsenio-gw-1#
pisa-unipi-arsenio-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.0.2.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 20 subnets, 3 masks
O       10.128.0.0/24 [110/1001] via 10.128.255.253, 00:02:31, Tunnel0
O       10.128.1.0/24 [110/1001] via 10.128.255.253, 00:02:31, Tunnel0
O       10.128.3.0/24 [110/1001] via 10.128.255.253, 00:02:31, Tunnel0
O       10.128.4.0/24 [110/1001] via 10.128.255.253, 00:02:31, Tunnel0
O       10.128.5.0/24 [110/1001] via 10.128.255.253, 00:02:31, Tunnel0
O       10.128.6.0/24 [110/1001] via 10.128.255.253, 00:02:31, Tunnel0
S       10.128.254.1/32 [1/0] via 10.128.255.253
C       10.128.254.5/32 is directly connected, Loopback0
O       10.128.255.0/30 [110/1001] via 10.128.255.253, 00:02:31, Tunnel0
O       10.128.255.4/30 [110/1001] via 10.128.255.253, 00:02:31, Tunnel0
O       10.128.255.8/30 [110/1002] via 10.128.255.253, 00:02:31, Tunnel0
C       10.128.255.252/30 is directly connected, Tunnel0
L       10.128.255.254/32 is directly connected, Tunnel0
O       10.129.0.0/24 [110/1002] via 10.128.255.253, 00:02:31, Tunnel0
O       10.129.1.0/24 [110/1002] via 10.128.255.253, 00:02:31, Tunnel0
O       10.129.128.0/24 [110/1003] via 10.128.255.253, 00:02:31, Tunnel0
O       10.130.0.0/24 [110/1002] via 10.128.255.253, 00:02:31, Tunnel0
O       10.130.1.0/24 [110/1002] via 10.128.255.253, 00:02:31, Tunnel0
C       10.131.0.0/24 is directly connected, FastEthernet0/0.401
L       10.131.0.1/32 is directly connected, FastEthernet0/0.401
     192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.0.2.0/24 is directly connected, FastEthernet0/1
L       192.0.2.20/32 is directly connected, FastEthernet0/1
S*   0.0.0.0/0 [1/0] via 192.0.2.1

pisa-unipi-arsenio-gw-1#
```

Выводы

Вывод

В ходе лабораторной работы была добавлена сеть Университета г. Пиза и выполнена её настройка в Cisco Packet Tracer.

Был создан GRE VPN-туннель между маршрутизаторами **pisa-unipi-arsenio-gw-1** и **msk-donskaya-arsenio-gw-1**.

Проверка с ноутбука администратора сети «Донская» показала успешную доступность узла **10.131.0.200** через туннель.